

Binary Spectral–Endpoint Bridges for the Fabius and Rvachev Functions

q-Pochhammer zero divisors, Thue–Morse endpoint weights, Mellin strip duality, and branch-resolved Lambert maps

Frontier research report based on a recursive audit of
Vladimir Reshetnikov’s `ProveIt/Analysis/FabiusFunction/docs` corpus

27 August 2026

Abstract

This report audits the recursive LaTeX corpus under `Analysis/FabiusFunction/docs` and then develops a new synthesis joining five strands that the existing documents largely treat in parallel: the Thue–Morse Mahler product, base-1/2 and base-1/4 *q*-calculus, the Rvachev infinite sinc product, the exact negative-Laplace decomposition at the Fabius endpoint, and the lower real branch of Lambert’s *W* function. The audit covers 57 .tex files: five living synthesis or reference documents, thirteen active frontier drafts, thirty-two frozen historical versions, and seven transcribed or corrected source papers. Frozen duplicate snapshots were checked through the archive’s provenance map and through the living documents into which their distinctive material was merged.

The principal new results, in the precise sense “not found explicitly in the inspected corpus”, are as follows. First, the Rvachev transform admits the *q*-Pochhammer product

$$\widehat{\text{up}}(z) = \prod_{\ell \geq 1} (z^2/\ell^2; 1/4)_\infty = \prod_{n \geq 1} (1 - z^2/n^2)^{v_2(n)+1},$$

so its integer zero multiplicity is exactly $v_2(n) + 1$. A finite version has both a Gaussian-binomial factorization and an exact signed Thue–Morse exponential-sum factorization. Second, if $\Theta(q) = \sum_{n \geq 0} (-1)^{\text{wt}_2(n)} q^n = \prod_{j \geq 0} (1 - q^{2^j})$, then the exponentially small endpoint correction in the negative-Laplace formula is exactly $-\log \Theta(e^{-s})$. Its Lambert-series coefficients satisfy

$$-\log \Theta(e^{-s}) = \sum_{n \geq 1} \frac{2^{v_2(n)+1} - 1}{n} e^{-ns} = \sum_{n \geq 1} \frac{2^{\text{ord}_z=n} \widehat{\text{up}} - 1}{n} e^{-ns},$$

giving a direct spectral–endpoint dictionary. Third, its Mellin transform is

$$\int_0^\infty [-\log \Theta(e^{-s})] s^{w-1} ds = \frac{\Gamma(w)\zeta(w+1)}{1 - 2^{-w}},$$

which realizes, on the complementary positive Mellin strip, the same meromorphic $\Gamma\zeta$ kernel used by the corpus for the negative-Laplace logarithm. This gives a new Thue–Morse/weighted-zero derivation of every Fourier coefficient of the endpoint’s log-periodic function and recovers the corpus’s already established exact mean in terms of the first Stieltjes constant. Fourth, the exact endpoint saddle is the fixed point of a periodically perturbed Lambert map, with the lower branch governing the small-argument endpoint regime and an explicit derivative proving local contraction there. Finally, an all-orders base-1/4 tail expansion places the corpus’s established centered moments in spectral tail coordinates and yields a certified acceleration of finite Thue–Morse approximants to the Fourier transform.

No assertion of global literature priority is made. Every result is labeled as established in the source corpus, newly proved here relative to that corpus, or conjectural/future work.

Contents

1	Scope, audit method, and proof-status discipline	4
1.1	What was inspected	4
1.2	Status labels	4
1.3	The structural gap that motivates the report	4
2	Established framework and normalization	5
2.1	The three functions	5
2.2	Probability model	5
2.3	Sinc product	6
2.4	Thue–Morse product	6
2.5	Negative Laplace transform and periodic correction	6
2.6	The Fourier-decay spectrum is not one number	7
3	A geometric q–Rvachev family	7
3.1	Definition and probabilistic realization	7
3.2	q –Pochhammer factorization	8
3.3	Cumulants in the q -family	9
4	The binary specialization: a 2-adic zero divisor	9
4.1	Finite triple factorization: sinc, q –Pochhammer, and Thue–Morse	10
5	The endpoint correction is a Thue–Morse logarithm	10
5.1	Exact identification	10
5.2	The 2-adic Lambert series	11
5.3	The spectral–endpoint dictionary	11
6	Mellin transform and the exact periodic phase	12
6.1	Mellin transform	12
6.2	Residues and the Fourier coefficients of P	13
6.3	The periodic correction to the saddle mean	14
7	Tilted cumulants and exact periodically perturbed Lambert saddles	14
7.1	Exact tilted mean	14
7.2	All tilted cumulants	14
7.3	Exact Lambert fixed point and branch threshold	15
7.4	Numerical check of the exact fixed point	16
8	All-orders sinc tails and centered Rvachev moments	16
8.1	Exact base-1/4 tail law	16
8.2	Centered cumulants and a Bernoulli–Mersenne recurrence	17
8.3	Certified Thue–Morse acceleration	18
9	How the new identities reorganize the existing theory	18
9.1	A commutative diagram of transforms	18
9.2	Two compatible q -geometries	19
9.3	Endpoint asymptotics from spectral data	19
9.4	Relation to the Fourier-decay audits	19

10 Further frontier problems	19
10.1 High-priority formalization targets	19
10.2 A general- q decay spectrum	20
10.3 Bridging the bases $1/2$ and $1/4$	20
10.4 A spectral derivation of the full endpoint expansion	20
10.5 Arithmetic nature of the periodic mean	20
10.6 Weighted zero statistics	21
11 Conclusions	21
A Supplementary derivations	21
A.1 The coefficient formula by partitions	21
A.2 A direct proof of the first-harmonic scale	22
A.3 The unperturbed and perturbed Lambert equations	22
B Recursive source inventory	22
B.1 Living documents: five files	22
B.2 Active frontier drafts: thirteen files	23
B.3 Frozen archive: thirty-two files	23
B.4 Source-paper transcriptions: seven files	24
C Notation crosswalk	24

1 Scope, audit method, and proof-status discipline

1.1 What was inspected

The repository separates formalization-backed exposition from non-formalized frontier mathematics. The living primary article records only statements with named Lean counterparts, whereas the frontier volume deliberately quarantines plausible or proved-on-paper results not yet represented by an exact formal declaration. The recursive source audit used that distinction rather than flattening all files into one undifferentiated literature review.

At the snapshot date, the recursive corpus contains the following groups.

Group	Files	Role and audit treatment
Living documents	5	Primary exposition, glossary, non-elementarity, Lean walkthrough, and frontier synthesis; used as the theorem and notation baseline.
Active drafts	13	Thue–Morse atlas and twelve Fourier-decay studies; checked for newer claims, conflicts, and unresolved questions.
Frozen archive	32	First versions and historical parallel drafts; checked through the archive provenance descriptions and merged-content map, with distinctive claims traced into living successors.
Source papers	7	Corrected or transcribed mathematical papers; used for provenance, comparison, and terminology.
Total	57	Recursive LaTeX corpus audited.

The full path-level inventory is recorded in [appendix B](#).

1.2 Status labels

Throughout the report:

- [E] means an established theorem or audited conclusion already present in the repository corpus (often formally verified in Lean);
- [N] means a theorem proved in this report and not located explicitly in the inspected corpus by path, phrase, and formula searches;
- [C] means a conjecture, program, or interpretation whose proof is not supplied.

“New” is therefore a corpus-relative statement, not an unsupported claim of first publication in the worldwide literature.

A final formula-by-formula overlap audit corrected three preliminary classifications before release. The exact mean and full Fourier spectrum of the endpoint-periodic function are already established in the corpus; so are the centered moment generating function, its initial rational moments, and a Bernoulli recurrence for those moments. Moreover, the same meromorphic $\Gamma\zeta$ kernel already occurs there as the Mellin transform of Λ on $-1 < \Re w < 0$. The new claims below are therefore restricted to the q -Pochhammer and weighted-zero dictionaries, the complementary positive-strip realization of that kernel, the resulting Thue–Morse/zero-divisor derivations, the exact Lambert branch map, and the certified finite-product consequences.

1.3 The structural gap that motivates the report

The corpus already develops, separately and deeply,

1. exact dyadic values of the Fabius function involving $(1/2; 1/2)_n$ and Gaussian binomial coefficients;
2. the Thue–Morse product $\prod_{j \geq 0} (1 - q^{2^j})$ and its Mahler equation;

3. the Rvachev Fourier transform as an infinite product of geometrically rescaled sinc functions;
4. a quadratic-plus-periodic negative-Laplace decomposition and a lower-Lambert endpoint phase;
5. a multi-regime theory of Fourier decay.

What is missing is an exact dictionary showing that the same binary arithmetic controls *both* the zero divisor of the Fourier transform and the exponentially small correction in the endpoint Laplace transform. Establishing that dictionary leads to a new q -Pochhammer zero-divisor formula, a weighted-zero interpretation of the corpus's $\Gamma\zeta$ Mellin kernel, exact tilted-cumulant recurrences, and branch-resolved Lambert- W maps.

Main synthesis in one line. The Thue–Morse product controls the endpoint correction; its logarithmic coefficients are $2^{\mu(n)} - 1$, where $\mu(n)$ is exactly the multiplicity of the Fourier zero at the integer n . Mellin transformation turns the resulting 2-adic Dirichlet series into the $\Gamma\zeta$ Fourier modes of the endpoint periodic function, and that periodic function perturbs the lower-Lambert saddle.

2 Established framework and normalization

2.1 The three functions

[E] The corpus carefully distinguishes three related global objects. The bounded Fabius function F is the distribution function on $[0, 1]$ satisfying

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x),$$

with constant extensions outside $[0, 1]$. Rvachev's even compactly supported bump is

$$\text{up}(x) = F(1 - |x|) \quad (|x| \leq 1), \quad \text{up}(x) = 0 \quad (|x| \geq 1).$$

The signed global extension $\mathcal{F}$ satisfies the pure dilation equation on all of $\mathbb{R}$ and should not be silently substituted for F or up .

2.2 Probability model

Let $U_1, U_2, \dots$ be independent $\text{Unif}[0, 1]$ variables and set

$$X = \sum_{j \geq 1} 2^{-j} U_j. \tag{1}$$

[E] Then $F(x) = \mathbb{P}(X \leq x)$. The centered variable

$$Y = 2X - 1 = \sum_{n \geq 0} 2^{-n} V_n, \quad V_n \stackrel{\text{iid}}{\sim} \text{Unif}[-1/2, 1/2], \tag{2}$$

has density up . Consequently Y is symmetric, $|Y| \leq 1$, and its Fourier transform under the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx$$

is a characteristic function on the real axis.

2.3 Sinc product

Write $\text{sinc } w = \sin w/w$ with the removable value 1 at $w = 0$, and set

$$\Phi(z) = \widehat{\text{up}}(z). \quad (3)$$

[E] The exact product and refinement law are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right), \quad \Phi(z) = \text{sinc}(\pi z) \Phi(z/2). \quad (4)$$

The product is entire and also admits the weighted cosine factorization $\Phi(z) = \prod_{m \geq 1} \cos(\pi z/2^m)^m$.

2.4 Thue–Morse product

Let

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}, \quad \Theta(q) = \sum_{n \geq 0} \varepsilon_n q^n. \quad (5)$$

[E] For $|q| < 1$,

$$\Theta(q) = \prod_{j \geq 0} (1 - q^{2^j}), \quad \Theta(q) = (1 - q) \Theta(q^2). \quad (6)$$

At finite depth,

$$\sum_{k=0}^{2^m-1} \varepsilon_k q^k = \prod_{j=0}^{m-1} (1 - q^{2^j}), \quad (7)$$

and Prouhet cancellation gives

$$\sum_{k=0}^{2^m-1} \varepsilon_k k^r = 0 \quad (0 \leq r < m), \quad \sum_{k=0}^{2^m-1} \varepsilon_k k^m = (-1)^m 2^{\binom{m}{2}} m!. \quad (8)$$

2.5 Negative Laplace transform and periodic correction

Define

$$B(s) = \mathbb{E} e^{-sX} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad \Lambda(s) = \log B(s), \quad s > 0. \quad (9)$$

[E] The exact dilation relation is

$$\Lambda(2s) = \Lambda(s) + \log(1 - e^{-s}) - \log s. \quad (10)$$

Let $L = \log 2$ and

$$T(s) = \sum_{m \geq 0} \log(1 - e^{-2^m s}), \quad \mathcal{E}(s) = -T(s) > 0. \quad (11)$$

Then

$$P(u) = \Lambda(2^u) + \frac{L}{2}(u^2 - u) + T(2^u) \quad (12)$$

is smooth and 1-periodic, and

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) + \mathcal{E}(s). \quad (13)$$

The corpus gives every Fourier mode of P , including its exact mean, and proves that no nonzero mode vanishes. It also obtains $\Gamma(w)\zeta(w+1)/(1-2^{-w})$ as the Mellin transform of Λ on $-1 < \Re w < 0$. One of our goals is to realize that same meromorphic kernel from the Thue–Morse endpoint tail on the complementary strip $\Re w > 0$, and thereby rederive the entire periodic spectrum by one weighted-zero residue calculation.

2.6 The Fourier-decay spectrum is not one number

[E] The comparative Fourier-decay audits establish a common lognormal core

$$\exp\left[-\frac{(\log x)^2}{2\log 2}\right]$$

but different power corrections for different notions of size. In the notation of the audits, the ordering is

$$\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0,$$

corresponding respectively to extremal rays, quadratic mean, Cesàro mean, and typical or almost-everywhere behavior. The second-wave audit additionally verifies that an unrestricted Cesàro normalization can be attained by a positive real-analytic decreasing gauge, although every stabilizing or bounded-distortion gauge fails. This report does not replace that multi-regime theory; instead, it supplies exact arithmetic and transform identities that sit underneath all regimes.

3 A geometric q -Rvachev family

3.1 Definition and probabilistic realization

The binary Rvachev product belongs naturally to a one-parameter family.

Definition 3.1 (q -Rvachev transform). For $0 < q < 1$, define

$$\Phi_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^{n/2} z). \quad (14)$$

Let $V_0, V_1, \dots$ be independent $\text{Unif}[-1/2, 1/2]$ variables and put

$$Y_q = \sum_{n \geq 0} q^{n/2} V_n. \quad (15)$$

Theorem 3.2 (Probability, support, smooth density, and refinement). *[N] The product (14) converges locally uniformly to an entire function. It is the Fourier transform of the law of Y_q , whose support is*

$$\left[-\frac{1}{2(1-\sqrt{q})}, \frac{1}{2(1-\sqrt{q})} \right].$$

The law has a compactly supported C^∞ density u_q , and

$$\Phi_q(z) = \text{sinc}(\pi z) \Phi_q(\sqrt{q} z). \quad (16)$$

Writing $a = \sqrt{q}$, the density satisfies

$$u_q(x) = \int_{(x-1/2)/a}^{(x+1/2)/a} u_q(y) dy, \quad (17)$$

$$u'_q(x) = \frac{1}{a} \left[u_q\left(\frac{x+1/2}{a}\right) - u_q\left(\frac{x-1/2}{a}\right) \right]. \quad (18)$$

At $q = 1/4$, $Y_q = Y$, $u_q = u$, and (18) is the usual Rvachev functional-differential equation.

Proof. The summands in (15) converge absolutely almost surely and are bounded by a geometric series. Independence gives (14). Since $\text{sinc}(\pi q^{n/2} z) = 1 + \mathcal{O}(q^n |z|^2)$ locally uniformly, the entire

product converges on compact sets. If $N \sim \log |x| / |\log \sqrt{q}|$, bounding the first N sinc factors by a constant times $(|x|q^{n/2})^{-1}$ gives

$$\log |\Phi_q(x)| \leq -\frac{(\log |x|)^2}{2|\log \sqrt{q}|} + \mathcal{O}(\log |x|),$$

so every polynomial multiple is integrable. Fourier inversion therefore yields a C^∞ density. Finally $Y_q \stackrel{d}{=} V_0 + aY'_q$ with an independent copy Y'_q . Convolving with the uniform density gives (17); differentiation gives (18). $\square$

3.2 q -Pochhammer factorization

Recall

$$(a; q)_m = \prod_{j=0}^{m-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j \geq 0} (1 - aq^j).$$

Theorem 3.3 (Euler- q factorization). *[N] For every $0 < q < 1$ and $z \in \mathbb{C}$,*

$$\Phi_q(z) = \prod_{\ell=1}^{\infty} (z^2/\ell^2; q)_\infty, \quad (19)$$

$$\Phi_{q,m}(z) := \prod_{n=0}^{m-1} \text{sinc}(\pi q^{n/2} z) = \prod_{\ell=1}^{\infty} (z^2/\ell^2; q)_m. \quad (20)$$

For $|z| < 1$,

$$\log \Phi_q(z) = -\sum_{r \geq 1} \frac{\zeta(2r)}{r(1-q^r)} z^{2r}, \quad (21)$$

$$\log \Phi_{q,m}(z) = -\sum_{r \geq 1} \frac{1-q^{mr}}{1-q^r} \frac{\zeta(2r)}{r} z^{2r}. \quad (22)$$

Proof. Euler's sine product gives

$$\text{sinc}(\pi q^{n/2} z) = \prod_{\ell \geq 1} \left(1 - \frac{q^n z^2}{\ell^2}\right).$$

The double product is absolutely and locally uniformly convergent because $\sum_{n,\ell} q^n/\ell^2 < \infty$, so the factors may be regrouped by ℓ . This proves (19) and (20). Expanding $\log(1-w) = -\sum_{r \geq 1} w^r/r$ and summing the geometric progression in n proves the logarithmic formulas. $\square$

Corollary 3.4 (Gaussian-binomial finite factorization). *[N] The finite product has the exact representation*

$$\Phi_{q,m}(z) = \prod_{\ell \geq 1} \left[\sum_{j=0}^m (-1)^j q^{\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_q \frac{z^{2j}}{\ell^{2j}} \right]. \quad (23)$$

Proof. Apply the finite q -binomial theorem $(a; q)_m = \sum_{j=0}^m (-1)^j q^{\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_q a^j$ to each factor in (20). $\square$

Remark 3.5. The living corpus uses $q = 1/2$ Gaussian coefficients in exact dyadic values of F . Equation (23) reveals a second, independent q -geometry: the spectral product naturally uses $q = 1/4$, the square of the spatial dilation. The two bases should not be conflated, but their coexistence is structural rather than accidental.

3.3 Cumulants in the q -family

Let $c_N(q) = \mathbb{E}Y_q^{2N}$, with $c_0(q) = 1$, and use the standard Bernoulli numbers B_{2r} .

Theorem 3.6 (q -Bernoulli cumulants and moments). *[N] All odd cumulants of Y_q vanish, and*

$$\kappa_{2r}(Y_q) = \frac{B_{2r}}{2r(1-q^r)}. \quad (24)$$

The even moments obey

$$c_N(q) = \frac{1}{2N} \sum_{r=1}^N \binom{2N}{2r} \frac{B_{2r}}{1-q^r} c_{N-r}(q). \quad (25)$$

Proof. Insert $z = t/(2\pi)$ in (21) and compare $\log \mathbb{E}e^{itY_q}$ with $\sum_{n \geq 1} \kappa_n(it)^n/n!$. Euler's evaluation of $\zeta(2r)$ gives (24). The standard moment-cumulant recursion $m_n = \sum_{k=1}^n \binom{n-1}{k-1} \kappa_k m_{n-k}$, followed by $\binom{2N-1}{2r-1} = \frac{r}{N} \binom{2N}{2r}$, gives (25). $\square$

4 The binary specialization: a 2-adic zero divisor

Set $q = 1/4$. Then $\Phi_{1/4} = \Phi = \widehat{\text{up}}$.

Theorem 4.1 (Arithmetic canonical product). *[N] The Rvachev transform has the genus-zero product*

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{\mu(n)}, \quad \mu(n) = v_2(n) + 1. \quad (26)$$

Thus the nonzero zeros of Φ are exactly the integers, and

$$\text{ord}_{z=n} \Phi = v_2(|n|) + 1 \quad (n \in \mathbb{Z} \setminus \{0\}). \quad (27)$$

Moreover

$$\sum_{n=1}^N \mu(n) = 2N - \text{wt}_2(N). \quad (28)$$

Proof. From (19),

$$\Phi(z) = \prod_{j \geq 0} \prod_{k \geq 1} \left(1 - \frac{z^2}{4^j k^2}\right).$$

For a fixed positive integer n , the equality $n = 2^j k$ with $k \in \mathbb{N}$ occurs exactly for $j = 0, \dots, v_2(n)$. Absolute convergence permits regrouping and proves (26). Each contributing factor has a simple zero, proving (27). Finally

$$\sum_{n \leq N} \mu(n) = N + \sum_{n \leq N} v_2(n) = N + v_2(N!) = 2N - \text{wt}_2(N)$$

by Legendre's formula. $\square$

Corollary 4.2 (Dirichlet series of the zero multiplicities). *For $\Re w > 1$,*

$$\sum_{n \geq 1} \frac{\mu(n)}{n^w} = \frac{\zeta(w)}{1 - 2^{-w}}. \quad (29)$$

Consequently (21) at $q = 1/4$ can be read directly from the zero divisor:

$$\log \Phi(z) = - \sum_{r \geq 1} \frac{z^{2r}}{r} \sum_{n \geq 1} \frac{\mu(n)}{n^{2r}} = - \sum_{r \geq 1} \frac{4^r \zeta(2r)}{r(4^r - 1)} z^{2r}. \quad (30)$$

Proof. Since $\mu(n) = \sum_{j \geq 0} \mathbf{1}_{2^j | n}$, Tonelli's theorem gives

$$\sum_n \mu(n) n^{-w} = \sum_{j \geq 0} \sum_{2^j | n} n^{-w} = \zeta(w) \sum_{j \geq 0} 2^{-jw}.$$

□

4.1 Finite triple factorization: sinc, q -Pochhammer, and Thue–Morse

Define

$$\Phi_m(z) = \prod_{n=0}^{m-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right), \quad S_m(z) = \sum_{k=0}^{2^m-1} \varepsilon_k e^{-2\pi i k z / 2^{m-1}}. \quad (31)$$

Theorem 4.3 (Finite triple factorization). *[N] For $m \geq 1$,*

$$\prod_{\ell \geq 1} (z^2 / \ell^2; 1/4)_m = \Phi_m(z) = \frac{2^{m(m-3)/2} e^{2\pi i z (1-2^{-m})}}{(i\pi z)^m} S_m(z). \quad (32)$$

The apparent singularity at $z = 0$ is removable. In fact S_m has a zero of exact order m there, with

$$S_m(z) = (2\pi i z)^m 2^{-m(m-1)/2} + \mathcal{O}(z^{m+1}). \quad (33)$$

Proof. The first equality is (20). Use $\sin w = e^{iw}(1 - e^{-2iw})/(2i)$ in each sinc factor. The powers of two give $2^{\sum_{n < m} n-m} = 2^{m(m-3)/2}$ and the phases sum to $e^{2\pi i z (1-2^{-m})}$. The remaining product is

$$\prod_{n=0}^{m-1} (1 - e^{-2\pi i z / 2^n}) = \prod_{j=0}^{m-1} (1 - w^{2^j}) = \sum_{k < 2^m} \varepsilon_k w^k, \quad w = e^{-2\pi i z / 2^{m-1}},$$

which is $S_m(z)$. Equation (8) annihilates the first m Taylor coefficients of S_m and gives the leading coefficient in (33). □

Corollary 4.4 (A Gaussian-binomial identity for a Thue–Morse exponential sum). *Combining (23) and (32) gives*

$$S_m(z) = (i\pi z)^m 2^{-m(m-3)/2} e^{-2\pi i z (1-2^{-m})} \times \prod_{\ell \geq 1} \left[\sum_{j=0}^m (-1)^j 4^{-\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_{1/4} \frac{z^{2j}}{\ell^{2j}} \right]. \quad (34)$$

This is an exact bridge between a finite signed Thue–Morse block and Gaussian coefficients at base $1/4$.

5 The endpoint correction is a Thue–Morse logarithm

5.1 Exact identification

Theorem 5.1 (Mahler identification of the Laplace tail). *[N] For every $s > 0$,*

$$e^{T(s)} = \Theta(e^{-s}) = \sum_{n \geq 0} \varepsilon_n e^{-ns}, \quad \mathcal{E}(s) = -\log \Theta(e^{-s}). \quad (35)$$

Consequently the exact periodic decomposition (13) becomes

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) - \log \Theta(e^{-s}). \quad (36)$$

Proof. Substitute $q = e^{-s}$ in the Thue–Morse product (6):

$$\Theta(e^{-s}) = \prod_{m \geq 0} (1 - e^{-2^m s}) = e^{T(s)}.$$

All factors are positive, so the real logarithm is unambiguous. The remaining formulas are immediate from (11) and (13). $\square$

5.2 The 2-adic Lambert series

Define

$$a(n) = \sum_{j=0}^{v_2(n)} 2^j = 2^{v_2(n)+1} - 1. \quad (37)$$

Theorem 5.2 (Lambert expansion and derivative hierarchy). *[N] For $s > 0$,*

$$\mathcal{E}(s) = \sum_{n \geq 1} \frac{a(n)}{n} e^{-ns}, \quad (38)$$

$$T^{(k)}(s) = (-1)^{k+1} \sum_{n \geq 1} a(n) n^{k-1} e^{-ns} \quad (k \geq 1). \quad (39)$$

In particular, with

$$R(s) = sT'(s), \quad (40)$$

one has

$$R(s) = \sum_{m \geq 0} \frac{2^m s}{e^{2^m s} - 1} = s \sum_{n \geq 1} a(n) e^{-ns}. \quad (41)$$

The beginning of the endpoint correction is

$$\mathcal{E}(s) = e^{-s} + \frac{3}{2}e^{-2s} + \frac{1}{3}e^{-3s} + \frac{7}{4}e^{-4s} + \frac{1}{5}e^{-5s} + \dots \quad (42)$$

Proof. Expand every logarithm in (11):

$$\mathcal{E}(s) = \sum_{m \geq 0} \sum_{k \geq 1} \frac{e^{-k2^m s}}{k}.$$

For a fixed n , the representations $n = k2^m$ are indexed by $m = 0, \dots, v_2(n)$, and $1/k = 2^m/n$. Their sum is $a(n)/n$. Termwise differentiation is justified by absolute convergence on every half-line $s \geq s_0 > 0$. $\square$

5.3 The spectral–endpoint dictionary

Theorem 5.3 (Endpoint coefficients from Fourier zero multiplicities). *[N] Let $\mu(n) = \text{ord}_{z=n} \Phi$. Then*

$$a(n) = 2^{\mu(n)} - 1, \quad (43)$$

and hence

$$-\log \Theta(e^{-s}) = \sum_{n \geq 1} \frac{2^{\text{ord}_{z=n} \Phi} - 1}{n} e^{-ns}. \quad (44)$$

Equivalently, the weighted zero-multiplicity Dirichlet series is

$$\mathfrak{Z}_\Phi(w) := \sum_{n \geq 1} \frac{2^{\text{ord}_{z=n} \Phi} - 1}{n^w} = \frac{\zeta(w)}{1 - 2^{1-w}}, \quad \Re w > 1. \quad (45)$$

Proof. By (27), $\mu(n) = v_2(n) + 1$, so (43) is (37). For the Dirichlet series, write $a(n) = \sum_{j \geq 0} 2^j \mathbf{1}_{2^j | n}$ and use Tonelli:

$$\sum_{n \geq 1} a(n) n^{-w} = \zeta(w) \sum_{j \geq 0} 2^{j(1-w)} = \frac{\zeta(w)}{1 - 2^{1-w}}.$$

□

Why this is a genuine bridge. The left side of (44) belongs to the small- x Fabius endpoint problem; the multiplicity on the right belongs to the zero divisor of the Rvachev Fourier transform. The equality does not merely compare asymptotic scales: it identifies every individual exponential coefficient with a local spectral multiplicity.

n	$v_2(n)$	$\text{ord}_{z=n} \Phi$	coefficient of e^{-ns} in $\mathcal{E}(s)$
1	0	1	1
2	1	2	3/2
3	0	1	1/3
4	2	3	7/4
6	1	2	1/2
8	3	4	15/8
12	2	3	7/12
16	4	5	31/16

6 Mellin transform and the exact periodic phase

6.1 Mellin transform

Theorem 6.1 (Positive-strip realization of the Gamma-zeta kernel). *[N] For $\Re w > 0$,*

$$\mathcal{ME}(w) := \int_0^\infty \mathcal{E}(s) s^{w-1} ds = \frac{\Gamma(w) \zeta(w+1)}{1 - 2^{-w}}. \quad (46)$$

Equivalently, the Mellin transform of $R(s) = sT'(s) = -s\mathcal{E}'(s)$ is

$$\int_0^\infty R(s) s^{w-1} ds = \frac{\Gamma(w+1) \zeta(w+1)}{1 - 2^{-w}}. \quad (47)$$

[E] The corpus obtains the same meromorphic right-hand side as the Mellin transform of Λ on the complementary fundamental strip $-1 < \Re w < 0$. Thus (46) is not a new special-function expression; its corpus-relative novelty is that the Thue–Morse endpoint tail and the weighted Fourier zero divisor realize that established kernel on $\Re w > 0$.

Proof. Put $\sigma = \Re w > 0$. Absolute convergence permits Fubini, because

$$\sum_{n \geq 1} \frac{a(n)}{n} \int_0^\infty e^{-ns} s^{\sigma-1} ds = \Gamma(\sigma) \sum_{n \geq 1} \frac{a(n)}{n^{\sigma+1}} < \infty.$$

Therefore (38) gives

$$\int_0^\infty \mathcal{E}(s) s^{w-1} ds = \Gamma(w) \sum_{n \geq 1} \frac{a(n)}{n^{w+1}}.$$

Use (45) with $w + 1$ in place of w . Integration by parts gives (47); the boundary terms vanish for $\Re w > 0$. □

6.2 Residues and the Fourier coefficients of P

Put

$$\chi_k = \frac{2\pi i k}{L}, \quad k \in \mathbb{Z}. \quad (48)$$

We use the Fourier convention $P(u) = \sum_{k \in \mathbb{Z}} \hat{P}(k) e^{2\pi i k u}$.

Theorem 6.2 (Mellin recovery of the full periodic spectrum). *[E] The periodic function in (13) has*

$$\boxed{\hat{P}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}, \quad k \neq 0,} \quad (49)$$

which agrees with the nonzero-mode formula already established in the corpus. Its mean is

$$\boxed{\hat{P}(0) = -\frac{L}{12} - \frac{1}{L} \left(\frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2} \right)}, \quad (50)$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant in the convention $\zeta(1+w) = w^{-1} + \gamma - \gamma_1 w + \mathcal{O}(w^2)$. Numerically,

$$\hat{P}(0) = -1.1090453654341866821907455547 \dots \quad (51)$$

These coefficient formulas, including the mean, are established in the corpus (the mean is formalized there). *[N] The corpus-relative novelty is the unified derivation below: both the oscillatory modes and the zero mode arise from the same Thue–Morse endpoint correction and the same meromorphic Mellin kernel.*

Proof. Mellin inversion writes $\mathcal{E}(s)$ as a vertical integral of

$$M(w)s^{-w}, \quad M(w) = \frac{\Gamma(w)\zeta(w+1)}{1-2^{-w}}.$$

The exponential decay of Γ on vertical lines justifies shifting the contour across $\Re w = 0$ and summing the resulting residues absolutely. Besides the pole at $w = 0$, the denominator has simple poles at $w = \chi_k$. The pole at $w = -\chi_k$ contributes

$$\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} s^{\chi_k} = \frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} e^{2\pi i k \log_2 s}.$$

On the other hand, (13) and $\Lambda(s) = \mathcal{O}(s)$ as $s \downarrow 0$ give

$$\mathcal{E}(s) = \frac{(\log s)^2}{2L} - \frac{1}{2} \log s - P(\log_2 s) + \mathcal{O}(s). \quad (52)$$

Comparison proves (49).

For the zero mode, expand at $w = 0$:

$$\begin{aligned} \Gamma(w)\zeta(1+w) &= \frac{1}{w^2} + c_0 + \mathcal{O}(w), & c_0 &= \frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2}, \\ \frac{1}{1-2^{-w}} &= \frac{1}{Lw} + \frac{1}{2} + \frac{Lw}{12} + \mathcal{O}(w^3). \end{aligned}$$

Hence

$$M(w) = \frac{1}{Lw^3} + \frac{1}{2w^2} + \left(\frac{L}{12} + \frac{c_0}{L} \right) \frac{1}{w} + \mathcal{O}(1).$$

The residue of $M(w)s^{-w}$ at zero is $(\log s)^2/(2L) - (\log s)/2 + L/12 + c_0/L$. Comparing with (52) shows that this constant is $-\hat{P}(0)$. $\square$

Remark 6.3 (Why the oscillation is easy to miss). The first nonzero mode is tiny:

$$|\hat{P}(1)| = 1.0627116252 \dots \times 10^{-6}, \quad |\hat{P}(2)| = 6.6927864 \dots \times 10^{-13}.$$

Differentiation magnifies the first mode, but it is still small. This explains why a non-periodic numerical fit can look excellent while being wrong at the sharp error scale.

6.3 The periodic correction to the saddle mean

Define

$$\Pi(u) = -\frac{1}{2} - \frac{P'(u)}{L}. \quad (53)$$

Then $\hat{\Pi}(0) = -1/2$, while for $k \neq 0$,

$$\hat{\Pi}(k) = -\frac{\Gamma(1 - \chi_k)\zeta(1 - \chi_k)}{L}. \quad (54)$$

The first harmonic has magnitude $9.6331836245 \dots \times 10^{-6}$.

7 Tilted cumulants and exact periodically perturbed Lambert saddles

7.1 Exact tilted mean

Under the exponentially tilted law

$$\frac{e^{-sX}}{B(s)} d\mathbb{P},$$

write $\mathbb{E}_s$ and Var_s for expectation and variance. Define

$$A(s) = -s\Lambda'(s) = s\mathbb{E}_s X. \quad (55)$$

Theorem 7.1 (Mean decomposition and dyadic recurrence). *[N] For every $s > 0$,*

$$A(s) = \log_2 s + \Pi(\log_2 s) + R(s). \quad (56)$$

It also satisfies the exact recurrence

$$A(2s) = A(s) + 1 - \frac{s}{e^s - 1}. \quad (57)$$

Proof. Apply $-s \frac{d}{ds}$ to (13); this gives (56). Differentiate (10) and multiply by $-s$ to obtain (57). $\square$

7.2 All tilted cumulants

Let

$$\kappa_n(s) = (-1)^n \Lambda^{(n)}(s), \quad K_n(s) = s^n \kappa_n(s), \quad D = s \frac{d}{ds}. \quad (58)$$

Thus $\kappa_n(s)$ is the n th cumulant of X under the tilted law.

Theorem 7.2 (Falling-Euler hierarchy). *[N] For $n \geq 1$,*

$$K_n(s) = (-1)^n (D)_{\underline{n}} \left[-\frac{L}{2} u(u-1) + P(u) - T(s) \right], \quad u = \log_2 s, \quad (59)$$

where $(D)_{\underline{n}} = D(D-1) \cdots (D-n+1)$. If $h(s) = \log(1 - e^{-s}) - \log s$, then

$$K_n(2s) = K_n(s) + (-1)^n s^n h^{(n)}(s). \quad (60)$$

In particular,

$$s^2 \text{Var}_s(X) = A(s) - DA(s) = u + \Pi(u) + R(s) - \frac{1 + \Pi'(u)}{L} - sR'(s) > 0, \quad (61)$$

and

$$K_2(2s) = K_2(s) + 1 - \frac{s^2 e^s}{(e^s - 1)^2}. \quad (62)$$

Proof. The operator identity $s^n \frac{d^n}{ds^n} = (D)_n$ proves (59). Differentiate (10) n times to obtain (60). Since $K_2 = s^2 \Lambda'' = A - DA$, substitution of (56) gives (61). Strict positivity follows because X is nondegenerate under every finite tilt. The explicit formula (62) follows by differentiating h twice. $\square$

7.3 Exact Lambert fixed point and branch threshold

The natural saddle for the inverse Laplace problem at $x \in (0, 1/2)$ is the unique solution of

$$x = -\Lambda'(s) = \mathbb{E}_s X. \quad (63)$$

Write $s = 2^u$ and

$$C(u) = \Pi(u) + R(2^u) = A(2^u) - u. \quad (64)$$

Then the exact saddle equation is

$$x2^u = u + C(u). \quad (65)$$

Theorem 7.3 (Periodically perturbed Lambert maps). *[N] Equation (65) has one solution $u = u_*(x)$ for every $0 < x < 1/2$. There is a unique $s_c > 0$ such that*

$$A(s_c) = \frac{1}{L}, \quad x_c := \frac{A(s_c)}{s_c} = \frac{1}{Ls_c}, \quad (66)$$

Numerically,

$$s_c = 3.5627054762160828049 \dots, \quad x_c = 0.4049436728688661444 \dots \quad (67)$$

For the endpoint range $0 < x < x_c$, the saddle is equivalently the fixed point of the lower-branch map

$$u = \mathfrak{L}_x^{(-1)}(u) := -\frac{1}{L} W_{-1} \left(-Lx 2^{-C(u)} \right) - C(u). \quad (68)$$

For $x_c < x < 1/2$, the same formula with W_0 in place of W_{-1} gives the saddle; at $x = x_c$ the two branches meet. At a fixed point of either branch,

$$\left(\mathfrak{L}_x^{(b)} \right)'(u) = \frac{C'(u)}{L(u + C(u)) - 1}, \quad b \in \{0, -1\}. \quad (69)$$

Because Π' is bounded and $R(2^u)$ is exponentially small as $u \rightarrow \infty$, the lower-branch map is a local contraction at $u_*(x)$ for all sufficiently small x .

Proof. The function $s \mapsto \mathbb{E}_s X$ is strictly decreasing because its derivative is $-\text{Var}_s(X) < 0$. Its limits are $1/2$ at $s \downarrow 0$ and 0 at $s \rightarrow \infty$, proving existence and uniqueness of the saddle. Also

$$A(s) = \sum_{j \geq 1} g(s/2^j), \quad g(t) = 1 - \frac{t}{e^t - 1},$$

and $g'(t) = (e^t(t-1) + 1)/(e^t - 1)^2 > 0$ for $t > 0$. Thus A increases continuously from 0 to infinity (for instance, the first $\lfloor \log_2 s \rfloor$ summands are uniformly positive once s is large), so (66) has one solution. Since $x = A(s)/s = \mathbb{E}_s X$ decreases with s , one has $A(s) > 1/L$ exactly when $x < x_c$.

Put $y = u + C(u) = A(2^u)$. Equation (65) becomes $ye^{-Ly} = xe^{-LC(u)}$. Multiplication by $-L$ selects W_{-1} when $y > 1/L$, W_0 when $0 < y < 1/L$, and the common branch value at $y = 1/L$. This proves the branch statements. Implicit differentiation of $ye^{-Ly} = xe^{-LC(u)}$, followed by subtraction of $C'(u)$, gives (69). In the endpoint regime the denominator tends to infinity, whereas C' stays bounded. $\square$

Corollary 7.4 (First correction to the unperturbed Lambert phase). *Let*

$$u_0(x) = -\frac{1}{L}W_{-1}(-Lx), \quad (70)$$

so that $x2^{u_0} = u_0$. Then, as $x \downarrow 0$,

$$u_*(x) = u_0(x) + \frac{C(u_0(x))}{Lu_0(x) - 1} + \mathcal{O}_C(u_0(x)^{-2}). \quad (71)$$

A single evaluation of the exact map (68) is typically much more accurate than the linear correction.

Proof architecture. Apply the mean-value theorem to $f(u) = x2^u - u - C(u)$ between u_0 and u_* . Since $f(u_0) = -C(u_0)$ and $f'(u) = Lx2^u - 1 - C'(u)$, the displacement is of order $1/u_0$. A second expansion gives (71); the implied constant depends only on uniform bounds for C and its first two derivatives. $\square$

7.4 Numerical check of the exact fixed point

The following values were computed at 70 decimal digits by summing the exact tilted mean

$$A(s) = \sum_{j \geq 1} \left(1 - \frac{s/2^j}{e^{s/2^j} - 1} \right)$$

and solving $x2^u = A(2^u)$.

x	u_0	u_*	linear-correction error	one-step map error
10^{-3}	13.7468091666470	13.6868068298588	1.3742×10^{-3}	7.4062×10^{-7}
10^{-6}	24.5491709694670	24.5175886975057	3.6440×10^{-4}	2.0539×10^{-7}
10^{-12}	45.3666989202317	45.3501796089457	9.7317×10^{-5}	6.5109×10^{-9}

The leading displacement is negative because $C(u) = -1/2 +$ a tiny periodic oscillation plus an exponentially small term.

8 All-orders sinc tails and centered Rvachev moments

8.1 Exact base-1/4 tail law

The finite product Φ_m from (31) satisfies

$$\frac{\Phi(z)}{\Phi_m(z)} = \Phi(z/2^m). \quad (72)$$

Theorem 8.1 (All-orders tail logarithm). *[N] If $|z| < 2^m$ and*

$$Q_m(z) = \frac{z^2}{4^{m-1}}, \quad (73)$$

then

$$\log \frac{\Phi(z)}{\Phi_m(z)} = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(4^r - 1)} Q_m(z)^r. \quad (74)$$

Writing

$$\frac{\Phi(z)}{\Phi_m(z)} = \sum_{N \geq 0} a_N Q_m(z)^N, \quad a_0 = 1, \quad (75)$$

the coefficients satisfy

$$Na_N = - \sum_{r=1}^N \frac{\zeta(2r)}{4^r - 1} a_{N-r}. \quad (76)$$

The first terms are

$$1 - \frac{\pi^2}{18}Q + \frac{19\pi^4}{16200}Q^2 - \frac{583\pi^6}{42865200}Q^3 + \frac{132809\pi^8}{1311675120000}Q^4 + \dots \quad (77)$$

Proof. Use (21) with $q = 1/4$ and replace z by $z/2^m$. The coefficient recurrence follows by logarithmic differentiation of (75). $\square$

8.2 Centered cumulants and a Bernoulli–Mersenne recurrence

Let

$$c_N = \int_{-1}^1 x^{2N} \text{up}(x) dx = \mathbb{E}Y^{2N}. \quad (78)$$

Theorem 8.2 (Tail coordinates for centered Rvachev moments). *[E] The corpus's centered law has cumulants*

$$\kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1}(Y) = 0, \quad (79)$$

and its centered moments c_N are already used in the exact dyadic theory. *[N] In the spectral coordinates of theorem 8.1, those moments satisfy the alternative Bernoulli–Mersenne recurrence*

$$c_N = \frac{1}{N} \sum_{r=1}^N \binom{2N}{2r} \frac{2^{2r-1} B_{2r}}{4^r - 1} c_{N-r}, \quad (80)$$

and the tail coefficient is exactly

$$a_N = \frac{(-1)^N c_N \pi^{2N}}{(2N)!}. \quad (81)$$

Proof. Specialize theorem 3.6 to $q = 1/4$. The recurrence is the corresponding specialization of (25). Finally

$$\Phi(z/2^m) = \sum_{N \geq 0} \frac{(-1)^N c_N}{(2N)!} \left(\frac{\pi z}{2^{m-1}} \right)^{2N},$$

and comparison with (75) gives (81). $\square$

The first moments are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}, \quad c_5 = \frac{46840699}{24405225075}. \quad (82)$$

The variance is $1/9$, and the fourth cumulant is $-2/225$.

Remark 8.3. The corpus already defines the centered moments, gives their hyperbolic-sine generating product, tabulates the values in (82), and proves a different Bernoulli recurrence. The new contribution here is narrower: the general- q derivation, the alternative cumulant-form recurrence (80), the exact identification of these established moments with the base- $1/4$ sinc-tail coefficients, and the certified Thue–Morse acceleration below.

8.3 Certified Thue–Morse acceleration

Combining the finite triple factorization with the moment tail gives an exact hybrid representation.

Theorem 8.4 (Thue–Morse spectral reconstruction). *[N] For every $m \geq 1$ and $z \in \mathbb{C}$,*

$$\begin{aligned} \Phi(z) &= \frac{2^{m(m-3)/2} e^{2\pi i z(1-2^{-m})}}{(i\pi z)^m} S_m(z) \\ &\quad \times \sum_{N=0}^{\infty} \frac{(-1)^N c_N}{(2N)!} \left(\frac{\pi z}{2^{m-1}} \right)^{2N}. \end{aligned} \quad (83)$$

For real z , truncation after $N = M$ has the certified error

$$\begin{aligned} \left| \Phi(z) - \Phi_m(z) \sum_{N=0}^M \frac{(-1)^N c_N}{(2N)!} \left(\frac{\pi z}{2^{m-1}} \right)^{2N} \right| \\ \leq \frac{(\pi|z|/2^{m-1})^{2M+2}}{(2M+2)!}. \end{aligned} \quad (84)$$

Proof. Multiply (32) by the moment series for $\Phi(z/2^m)$. For real z , interpret the tail as $\mathbb{E} \cos((\pi z/2^{m-1})Y)$ and use the Lagrange remainder for cosine together with $|Y| \leq 1$ and $|\Phi_m(z)| \leq 1$. $\square$

Example 8.5 (Numerical acceleration). At $z = 10$ and $m = 6$, the exact tail is $\Phi(10/64) = 0.94753140589767634644 \dots$. The errors after retaining M moment corrections, together with the rigorous bound (84), are

M	actual tail error	certified upper bound
0	5.24686×10^{-2}	4.81914×10^{-1}
1	1.07744×10^{-3}	3.87069×10^{-2}
2	1.20907×10^{-5}	1.24356×10^{-3}
3	8.69394×10^{-8}	2.14032×10^{-5}

Thus a finite signed Thue–Morse sum, corrected by only a few rational moments, gives a rapid and rigorously controlled evaluation of the infinite sinc product.

9 How the new identities reorganize the existing theory

9.1 A commutative diagram of transforms

The central relationships can be read as the following chain:

$$\begin{array}{ccccc} \Theta(q) = \prod (1 - q^{2^j}) & \xrightarrow{q=e^{-s}} & \mathcal{E}(s) = -\log \Theta(e^{-s}) & \xrightarrow{\mathcal{M}} & \frac{\Gamma(w)\zeta(w+1)}{1-2^{-w}} \\ & & \updownarrow & & \downarrow \text{residues} \\ \Phi(z) = \prod \text{sinc}(\pi z/2^j) & \xrightarrow{\text{zero divisor}} & \mu(n) = v_2(n) + 1 & \xrightarrow{a(n)=2^{\mu(n)}-1} & P(\log_2 s). \end{array}$$

The periodic endpoint function is therefore not an unrelated analytic decoration. It is the residue shadow of a Thue–Morse logarithm whose coefficients can be read from the Rvachev spectral zero divisor.

9.2 Two compatible q -geometries

The corpus’s exact dyadic arithmetic uses base $q = 1/2$ because each spatial refinement changes the dyadic denominator by one power of two. The Fourier product uses base $q = 1/4$ because its Taylor and Euler products are even in z , so the same spatial dilation becomes a squared spectral dilation. This explains why

- $(1/2; 1/2)_n$ and $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}$ arise in exact Fabius values;
- $(z^2/\ell^2; 1/4)_\infty$, $4^r - 1$, and $\left[\begin{smallmatrix} m \\ j \end{smallmatrix}\right]_{1/4}$ arise in the sinc product and centered moments.

A future theory should treat these not as competing normalizations but as the arithmetic and spectral faces of the same binary refinement.

9.3 Endpoint asymptotics from spectral data

The exact identity (44) gives a new route toward the endpoint expansion. Starting only with the Fourier zero multiplicities,

1. form $a(n) = 2^{\mu(n)} - 1$;
2. build $\mathcal{E}(s) = \sum a(n)e^{-ns}/n$;
3. Mellin transform to obtain (46);
4. read off the quadratic logarithm and all periodic modes;
5. insert them into the exact saddle map (68).

This does not yet derive every coefficient of the all-orders Fabius endpoint expansion, but it supplies the entire oscillatory datum and the exact periodic saddle displacement from the spectral zero divisor alone.

9.4 Relation to the Fourier-decay audits

The arithmetic product (26) is not by itself a pointwise asymptotic for $|\Phi(x)|$: near integers, zero multiplicities dominate, while extremal rays avoid zeros in highly structured ways. It nevertheless offers three tools for the audited decay problem:

1. exact zero multiplicities and cumulative zero counting;
2. a q -family in which the binary ratio can be varied continuously;
3. certified finite-shell corrections suitable for rigorous numerics and Lean formalization.

The separation of $\kappa_\infty, \kappa_2, \kappa_1, \kappa_0$ remains essential.

10 Further frontier problems

10.1 High-priority formalization targets

The following statements appear especially suitable for Lean because their proofs reduce to finite products, absolutely convergent series, or elementary operator identities.

1. **Arithmetic canonical product:** formalize $\Phi(z) = \prod_{n \geq 1} (1 - z^2/n^2)^{v_2(n)+1}$ and the integer zero multiplicity.
2. **Finite triple factorization:** combine the finite sinc product with the formal Thue–Morse polynomial and Prouhet cancellation.

3. **Endpoint Mahler identity:** prove $\mathcal{E}(s) = -\log \Theta(e^{-s})$ and the 2-adic Lambert expansion.
4. **Spectral-endpoint coefficient theorem:** identify the Lambert coefficient with $2^{\text{ord}_n \Phi} - 1$.
5. **Centered moment recurrence:** derive (80) from the already formal sinc product.
6. **Periodic Lambert map:** package the exact saddle equation and derivative (69), using strict tilted variance for uniqueness.

The Mellin residue computation and Stieltjes-constant mean are analytically heavier but can be isolated after the elementary core is formalized.

10.2 A general- q decay spectrum

Conjecture 10.1 (Smooth dependence of decay exponents). *[C] For the geometric product Φ_q , the analogues of the extremal, L^2 , Cesàro, and typical power exponents exist for every $0 < q < 1$ and vary continuously, perhaps real-analytically away from resonance parameters at which zero lattices overlap.*

The universal quadratic coefficient should be $1/(2|\log \sqrt{q}|) = 1/|\log q|$, as the elementary upper bound in [theorem 3.2](#) already suggests. The lower-order exponents should arise from transfer operators depending on q .

10.3 Bridging the bases $1/2$ and $1/4$

Conjecture 10.2 (Arithmetic-spectral q -transform). *[C] There is a direct coefficient transform carrying the base- $1/2$ Gaussian-binomial formulas for exact dyadic values of F to the base- $1/4$ Bernoulli–Mersenne moments of up , without passing through the full probability model.*

A likely starting point is the identity $Y = 2X - 1$, followed by a binomial transform of the half moments, but a satisfactory formula should preserve the q -Pochhammer structure rather than expand it away.

10.4 A spectral derivation of the full endpoint expansion

Conjecture 10.3 (Zero-divisor endpoint reconstruction). *[C] The complete all-orders endpoint asymptotic of $F(x)$, including every periodic coefficient, can be reconstructed from the weighted zero zeta function $\mathfrak{Z}_\Phi$ together with the local normalization $\Phi(0) = 1$.*

The report reconstructs the quadratic logarithm and rederives the entire periodic function P , including its established mean, from the weighted zero data; it also derives the exact saddle displacement. The remaining challenge is to derive the full Gaussian saddle-jet algebra directly from spectral data.

10.5 Arithmetic nature of the periodic mean

Formula (50) introduces the combination

$$\frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2}.$$

Its arithmetic nature is unknown. Even without transcendence information, the exact formula is useful: it separates the mean from the exponentially tiny nonzero modes and eliminates one numerical constant from the endpoint theory.

10.6 Weighted zero statistics

The ordinary multiplicity and endpoint-weighted multiplicity have Dirichlet series

$$\sum \frac{\mu(n)}{n^w} = \frac{\zeta(w)}{1 - 2^{-w}}, \quad \sum \frac{2^{\mu(n)} - 1}{n^w} = \frac{\zeta(w)}{1 - 2^{1-w}}.$$

The second has a double singularity at $w = 1$ after multiplication by the pole of ζ , which is the arithmetic source of the quadratic logarithm in the endpoint correction. A Tauberian development of its summatory function, with explicit binary periodic fluctuation, should give an elementary real-variable route to the Mellin result.

11 Conclusions

The inspected corpus already contains an unusually complete and carefully status-separated theory of the Fabius and Rvachev functions. The main contribution of this report is not a new isolated formula but a new exact dictionary:

1. the infinite sinc product is a base-1/4 q -Pochhammer product;
2. its integer zero multiplicity is $v_2(n) + 1$;
3. the Thue–Morse endpoint logarithm weights that multiplicity by $2^\mu - 1$;
4. the resulting Dirichlet series has Mellin transform $\Gamma(w)\zeta(w+1)/(1 - 2^{-w})$;
5. the residues reproduce the endpoint's full established log-periodic spectrum, including its formalized mean, through a new Thue–Morse/Mellin route;
6. the same periodic function enters an exact lower- W_{-1} fixed point for the saddle;
7. finite Thue–Morse sums, corrected by centered Bernoulli–Mersenne moments, give certified approximations to the full Fourier product.

This unifies the Thue–Morse, q -binomial, sinc-product, endpoint, and Lambert- W themes without erasing their different normalizations. It also leaves a concrete formalization program and several testable frontier questions.

A Supplementary derivations

A.1 The coefficient formula by partitions

From the cumulant generating function, the centered moments also admit the explicit partition sum

$$c_N = (2N)! \sum_{\substack{m_1, \dots, m_N \geq 0 \\ \sum_{r=1}^N r m_r = N}} \prod_{r=1}^N \frac{1}{m_r!} \left[\frac{2^{2r-1} B_{2r}}{r(4^r - 1)(2r)!} \right]^{m_r}. \quad (85)$$

This makes rationality immediate and displays the product of Mersenne-type factors $4^r - 1$. Since

$$\prod_{r=1}^N (4^r - 1) = 4^{N(N+1)/2} (1/4; 1/4)_N,$$

the denominator geometry is again governed by a base-1/4 q -Pochhammer symbol.

Post-snapshot Lean status. The universal algebraic shape of (85) is now formalized in `ExponentialPartition.lean`: with

$$E_r = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)(2r)!},$$

`partitionExpSum_eq_sum_div` identifies `partitionExpSum E N` with the displayed partition sum before multiplication by $(2N)!$. `ExponentialBell.lean` further identifies that coefficient function with `SaddleExpansion.expCoeff`. These are generic coefficient identities only: the cumulant formula, its identification with c_N , the specialized Bernoulli–Mersenne recurrence, and a standalone complete-Bell-polynomial interface remain separate formal obligations. The [N] classifications above therefore remain unchanged.

A.2 A direct proof of the first-harmonic scale

For $k \neq 0$, Stirling’s formula on vertical lines gives

$$|\Gamma(-\chi_k)| \asymp |\chi_k|^{-1/2} e^{-\pi|\chi_k|/2}.$$

Since $|\chi_1| = 2\pi/L \approx 9.0647$, the gamma factor alone contributes an exponential suppression near $e^{-14.24}$. The zeta factor is only of polynomial size. This explains the extraordinary numerical smallness of the periodic modes; their nonvanishing follows separately from the zero-free line $\Re s = 1$ for the Riemann zeta function.

A.3 The unperturbed and perturbed Lambert equations

The familiar lower-Lambert phase solves

$$x2^{u_0} = u_0, \quad u_0 = -L^{-1}W_{-1}(-Lx).$$

The exact phase replaces u_0 on the right by $u + C(u)$. The map (68) is not an ad hoc iteration: it solves the Lambert part exactly while freezing only the bounded periodic perturbation. Its derivative (69) is therefore of order $1/u$, explaining the rapid numerical convergence seen in the table.

B Recursive source inventory

B.1 Living documents: five files

1. `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`
2. `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`
3. `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`
4. `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`
5. `non-formalized-research-frontiers/non-formalized-research-frontiers.tex`

B.2 Active frontier drafts: thirteen files

- Atlas file: `non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex`.
- Twelve one-file directories under `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/`:
 - `Rvachev_Up_Fourier_Decay-2/`
 - `Rvachev_Up_Fourier_Decay-3/`
 - `Rvachev_Up_Fourier_Decay_Comparative_Audit/`
 - `Rvachev_Up_Fourier_Decay_Gentle_Guide/`
 - `Rvachev_Up_Fourier_Decay_Second_Wave_Audit/`
 - `asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/`
 - `rvachev_up_fourier_decay-1/`
 - `rvachev_up_fourier_decay-4/`
 - `rvachev_up_fourier_decay-5/`
 - `rvachev_up_fourier_decay-6/`
 - `rvachev_up_fourier_decay-7/`
 - `rvachev_up_fourier_decay-8/`

Together these are one Atlas file plus twelve Fourier-decay files.

B.3 Frozen archive: thirty-two files

The archive contains one historical `.tex` per entry. Its README states that byte-identical duplicates were pruned and that the retained entries are frozen first or special historical versions. The entries are:

Archive group	Entry
Early notes	<code>FabiusAsymptotic; K-foldsummationoverthesignedThue-Morsesequence</code>
Standalone studies	<code>Non_Elementarity_of_the_Fabius_Function; Small_Argument_Asymptotics; fabius-inverse-dyadic-closed-form</code>
Primary exposition	<code>Fabius_Function_and_Rvachev_Up; Fabius_Function_and_Rvachev_Up-2; Fabius_Function_and_Rvachev_Up-3; Fabius_and_Rvachev_Up-2; Fabius_and_Rvachev_up; Fabius_function_and_up_function</code>
Primary supplements	<code>Fabius_Function_Missing_Parts; Fabius_Function_Missing_Parts-2</code>
q -calculus	<code>Fabius_q_Special_Functions; fabius-q-special-functions</code>
Lean walkthrough	<code>Fabius_Function_Lean_Walkthrough; fabius_lean_walkthrough</code>
Glossary	<code>FabiusFunction_Mathematical_Glossary; FabiusFunction_Mathematical_Glossary-2</code>

Archive group	Entry
Research frontiers	Fabius_Dyadic_Asymptotic_Bridge; Fabius_Dyadic_Formulae_and_Alternative_Representations; Fabius_Dyadic_Formulae_to_Asymptotics; Fabius_Dyadic_q_Connections; Fabius_Thue_Morse_Convergence_Rate; Fabius_Thue_Morse_Convergence_Rates; Primary_Exposition_Gap_Register; Repeated_Integration_and_Rvachev_Up; Rvachev_Up_from_Repeated_Integration; non-formalized-research-frontiers; Thue_Morse_Formula_Atlas-1; Thue_Morse_Formula_Atlas-2; Thue_Morse_Formula_Atlas-3

B.4 Source-paper transcriptions: seven files

1. AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
2. AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
3. Lacunary_products/Lacunary_products.tex
4. 09-Function/09-Function.tex
5. 157-Arithmetic-v3/157-Arithmetic-v3.tex
6. document/document.tex
7. ubiq15/ubiq15-corrected.tex

C Notation crosswalk

Symbol	Meaning
F	bounded Fabius distribution function
up	Rvachev's even compactly supported density
X	$\sum_{j \geq 1} 2^{-j} U_j$, supported on $[0, 1]$
Y	$2X - 1$, with density up
Φ	$\widehat{\text{up}}$, the binary infinite sinc product
ε_n	signed Thue–Morse value $(-1)^{\text{wt}_2(n)}$
$\Theta(q)$	Thue–Morse Mahler product $\sum \varepsilon_n q^n$
B, Λ	negative Laplace transform of X and its logarithm
$T, \mathcal{E}$	forward logarithmic tail and positive endpoint correction
P	smooth one-periodic term in the exact Laplace decomposition
R	$sT'(s)$, the exponentially small derivative correction
A	$-s\Lambda'(s) = s\mathbb{E}_s X$
Π	$-1/2 - P'/L$, the periodic part of $A - \log_2 s$
C	$\Pi(u) + R(2^u)$, the exact bounded saddle perturbation
$\mu(n)$	integer-zero multiplicity $\text{ord}_{z=n} \Phi = v_2(n) + 1$

References

- [1] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, ProveIt repository, living exposition, snapshot 27 August 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up
- [2] V. Reshetnikov, *Non-formalized Research Frontiers for the Fabius Function*, ProveIt repository, living frontier synthesis, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers>
- [3] V. Reshetnikov, *Thue–Morse Formula Atlas*, active draft, ProveIt repository, 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas
- [4] V. Reshetnikov, *The Decay Envelope of the Rvachev Sinc Product: Comparative Audit*, active draft, ProveIt repository, 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit
- [5] V. Reshetnikov, *The Decay Envelope of the Rvachev Sinc Product: Second-Wave Audit*, active draft, ProveIt repository, 27 August 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit
- [6] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation of the 1982 article; transcription included in the ProveIt documentation corpus.
- [7] J. Arias de Reyna, *Arithmetic of the Fabius function*, transcription included in the ProveIt documentation corpus.
- [8] J.-P. Allouche and J. Shallit, *The ubiquitous Prouhet–Thue–Morse sequence*, source-paper transcription included in the ProveIt corpus.
- [9] C. Aistleitner, M. Hofer, and G. Larcher, *On evil Kronecker sequences and lacunary trigonometric products*, *Annales de l’Institut Fourier* **67** (2017), 637–687; corrected transcription included in the ProveIt corpus.
- [10] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, *On the Lambert W function*, *Advances in Computational Mathematics* **5** (1996), 329–359.